

US Road Accidents

Analysis & Prediction

Project Team

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Chapter 1

Project Overview

*Concept · Data Source · Project Impact · SMART
Questions*

1. Project Concept

This project focuses on analyzing U.S. traffic accident data covering 500,000 records from 2016 to 2023, sourced from a nationwide Kaggle dataset spanning 49 states. The goal is to uncover critical safety patterns related to weather conditions, geographic location, time of day, and road infrastructure. Using a Medallion data architecture (Bronze → Silver → Gold → ML layers) built on Microsoft Fabric, the project delivers a full data engineering pipeline, interactive Power BI dashboards, time-series forecasting with Prophet, and a production-grade machine learning model (CatBoost V4) for accident severity prediction. The entire system provides data-driven insights to support road safety decision-making and emergency response planning.

2. Data Source

The dataset used in this project was sourced from Kaggle and represents a comprehensive nationwide collection of U.S. traffic accidents spanning multiple years.

Attribute	Detail
Source	Kaggle (publicly available dataset)
Geographic Coverage	49 states across the USA
Time Coverage	2016 – 2023 (87 months)
Total Records (after cleaning)	~500,000 accident records
Data Points per Record	Location, time, weather conditions, road attributes, severity level

1.3 Project Impact

The analysis and modeling of this dataset provide critical insights that can be utilized across various sectors:

- **Risk Hotspot Identification:** Identifying the highest-risk states and cities to help authorities prioritize road safety improvements. California alone accounts for 113K accidents (22.6% of the total dataset).
- **Weather Impact Analysis:** Understanding how different environmental conditions (e.g., rain, fog, snow) correlate with accident frequency and severity. Notably, 94.75% of accidents occur in normal weather — fair, cloudy, and clear conditions — which challenges common assumptions.
- **Emergency Service Optimization:** Providing data-driven insights to help optimize the distribution and response times of emergency services, especially during identified high-risk windows (Friday rush hour, December peak).
- **Insurance & Actuarial Support:** Assisting insurance companies in refining risk assessment models based on location, time, and environmental patterns.
- **Severity Prediction:** Building a production-grade Machine Learning model (CatBoost V4) to predict accident severity with a ROC-AUC of 0.746, enabling proactive safety measures and better resource management.

1.4 Main Charts

The Power BI dashboard is built around eight core visualization types, each chosen to answer a specific analytical question:

Chart	Purpose
Map by State	Shows where accidents are geographically concentrated across the USA
Severity Distribution	Breaks down accidents into Low, Moderate, High, and Critical Impact categories
Year-over-Year Trend	Shows how total accident counts changed from 2016 to 2023

Chart	Purpose
Monthly Trend	Reveals seasonal and monthly patterns in accident volume
Weather Impact Charts	Links weather conditions (rain, fog, clear) to accident frequency
Road Feature Charts	Analyses the effect of traffic signals, junctions, and crossings
Model Accuracy Comparison	Compares the performance of all trained ML models' side by side
12-Month Forecast	Projects expected accident volume for the next 12 months using Prophet

1.5 SMART Questions

The project was designed to answer ten specific, measurable questions about U.S. road safety:

#	Question
1	Which U.S. states have the highest number of traffic accidents?
2	When do accidents occur most frequently by hour, day, and month?
3	Which weather conditions are associated with the highest number of accidents?
4	Does low visibility increase accident frequency and severity?
5	Which cities have the highest accident volume?
6	Do traffic signals, junctions, and crossings affect accident occurrence?
7	Which locations recorded the highest percentage of severe accidents?
8	Can machine learning accurately predict accident severity?
9	What are the top factors influencing accident severity?
10	Did accident volume increase or decrease compared with previous years?

Chapter 2

Technical Architecture

Tools · Medallion Architecture · Data Model

2.1 Tools Used

The project was built on a modern data engineering stack, combining cloud infrastructure, Python-based processing, and leading business intelligence tools:

Category	Tools / Technologies
Cloud & Infrastructure	Microsoft Fabric (OneLake, Lakehouse, SQL Endpoint)
Programming Language	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, XGBoost, LightGBM, CatBoost, Prophet
Visualization	Power BI, Matplotlib, Seaborn
UI/UX Design	Figma

2.2 Architecture (Medallion)

The project uses a Medallion Architecture — a layered data pipeline that progressively refines raw data into analysis-ready datasets. Think of it like purifying water through multiple filters, each making it cleaner and more useful than the last.



Layer	What It Contains	Description
Bronze	Raw CSV files ingested as-is	Raw CSV files ingested into OneLake as-is. No transformations applied — preserves the original source data.
Silver	Cleaned & standardised data	Data cleaned, standardized, and validated. Missing values handled, data types corrected, and date/time features prepared.
Gold	Star Schema for BI (1 Fact + 7 Dimension tables)	1 fact table + 7 dimension tables designed for scalable Power BI reporting. Optimized for aggregation and filtering.
ML Layer	Feature Engineering + Model Predictions	Additional derived features (rush hour flag, cyclical time encoding, visibility category) added. ML models trained and predictions stored here.

2.3 Data Model

 [View Prediction Service](#)

The Gold layer is structured as a Star Schema — the industry standard for business intelligence. It contains one central Fact Table surrounded by seven Dimension Tables.

What is a Star Schema?

A Star Schema places all measurable data (accidents, severity scores, durations) in a central Fact Table, and all descriptive data (location, time, weather, road conditions) in surrounding Dimension Tables. This design makes reporting extremely fast and flexible — you can slice and filter data in any combination without complex joins.

Table	Type	Contains
fact_accidents	Fact	Core accident records: ID, severity, duration, location key, time key, weather key

Table	Type	Contains
dim_location	Dimension	State, city, street, latitude, longitude
dim_time	Dimension	Year, month, day, hour, day of week, rush hour flag
dim_weather	Dimension	Weather condition, temperature, visibility, wind speed, humidity
dim_road	Dimension	Traffic signal, junction, crossing, road feature count
dim_severity	Dimension	Severity label (Low / Moderate / High / Critical), severity score
dim_sunrise	Dimension	Day or Night indicator (Sunrise/Sunset)
ml_predictions	ML Output	CatBoost risk score, predicted risk level (High/Low), threshold used

Chapter 3

Data Engineering & Data analysis

*Cleaning Process · Feature Engineering · ML
Models*

3.1 Cleaning Process

Before any analysis or modelling could begin, the raw accident data had to be thoroughly cleaned and standardized. Raw data from Kaggle often contains inconsistencies, missing values, and formatting issues. The following steps were applied:

Cleaning Step	What Was Done	Why It Matters
Missing Value Imputation	Used domain-specific logic (e.g., fill weather from nearby records, median for numeric columns)	Numerical columns (Temperature_F, Wind_Speed_mph) imputed using median values grouped by state and month. Boolean road feature flags filled as False where absent.
Zero Rows Dropped	No records were deleted — all 500K+ entries were preserved	All 500,000 records were preserved to maintain statistical representativeness.
Column Standardization	Renamed columns for consistency, converted data types (e.g., strings → datetime)	Column names normalized (snake_case), data types cast to correct formats (datetime, float, boolean, categorical), duplicate records removed.
Date/Time Feature Prep	Parsed start and end timestamps into Year, Month, Hour, Day of Week	Year, Month, Day, Hour, Day_of_Week, and Week_of_Year columns derived to support all temporal analysis.
Cross-Layer Validation	Each layer (Bronze → Silver → Gold) was validated for row counts, nulls, and data ranges	Each Silver-to-Gold transformation included checks to ensure record counts, severity distributions, and KPI totals remained consistent.

3.2 Feature Engineering

Feature Engineering is the process of creating new, more informative variables from existing data. These engineered features often improve model accuracy significantly because they capture patterns that raw data alone cannot express.

Feature	How It Was Created	What It Tells the Model
Year / Month / Hour / Week_of_Year	Extracted from Start_Time timestamp	Captures seasonal patterns, rush hour effects, and year-over-year trends
Rush_Hour Flag	1 if Hour is 7–9 AM or 3–6 PM on a weekday, else 0	Identifies accidents during peak commuting times — a major risk factor
Weather_Type Group	Grouped 100+ weather conditions into 'Normal' or 'Extreme'	Simplifies weather into actionable risk categories without losing signal
Visibility_Category	Binned visibility (miles) into Low / Moderate / High	Low visibility is a known accident risk — this makes it usable as a feature
Road_Features_Count	Counted how many road features (signal, junction, crossing, etc.) are present	More road features = more complex intersection = potentially higher risk
Cyclical Encoding	Encoded Hour and Month using sin/cos transformation	Prevents the model from thinking Hour 23 and Hour 0 are far apart — they are adjacent

Chapter 4

Analysis & Machine learning

*Models · Results · Model Selection · Feature
Importance*

4.1 ML Models Overview

The machine learning objective was to classify each accident into one of two risk categories: High Risk (severity 3–4) or Low Risk (severity 1–2). This binary classification task was tested across five model families, each trained in both binary and multi-class configurations on 20 engineered features from the Gold + ML layer.

Model	Type	Notes
Logistic Regression	Linear	Baseline to establish a performance floor.
Random Forest	Ensemble	Strong general-purpose baseline.
XGBoost	Gradient Boosting	Gradient boosting with regularization; widely used for tabular data.
LightGBM	Gradient Boosting	Fast gradient boosting with leaf-wise tree growth; strong on large datasets.
CatBoost V4	Gradient Boosting	Native categorical feature handling and ordered boosting. Selected as production model.

4.2 Model Results & Comparison

- **Binary Classification:** Predict whether an accident is High Risk (Severity 3–4) or Low Risk (Severity 1–2). This is easier for a model to learn and more actionable for emergency response.
- **Multi-class Classification:** Predict the exact severity level (1, 2, 3, or 4). More granular but harder to learn due to class imbalance.

Model	Task	Accuracy	ROC-AUC	Production?	Notes
CatBoost V4	Binary	69.76%	74.60%	✓ YES	Best real-world performance — genuinely separates High vs Low Risk

Model	Task	Accuracy	ROC-AUC	Production?	Notes
LightGBM	Binary	78.62%	50.00%	✗ No	High accuracy but ROC-AUC = 0.50 means it cannot distinguish risk classes
XGBoost	Binary	76.85%	50.00%	✗ No	Same issue as LightGBM — high accuracy by predicting majority class only
Random Forest	Binary	68.10%	50.10%	✗ No	Moderate performance, no meaningful class separation
Logistic Regression	Binary	50.06%	49.90%	✗ No	Linear model — too simple for this non-linear problem
LightGBM	Multi-class	69.83%	—	✗ No	Best multi-class accuracy — used for multiclass benchmarking
XGBoost	Multi-class	68.34%	—	✗ No	Second best multi-class
Random Forest	Multi-class	66.41%	—	✗ No	Strong baseline for multi-class
Logistic Regression	Multi-class	43.94%	—	✗ No	Linear baseline — as expected, underperforms on complex data

Why CatBoost Was Chosen Over LightGBM

LightGBM has 78.62% accuracy but a ROC-AUC of only 0.50 — this means it predicts the majority class (Low Risk) almost every time. It looks accurate but is useless for identifying dangerous accidents. CatBoost has a lower raw accuracy (69.76%) but a ROC-AUC of 0.746, which means it genuinely learns to separate High Risk from Low Risk cases. Its decision threshold was also optimised to 0.5034 to balance precision and recall. In real-world terms: CatBoost correctly flags dangerous accidents. LightGBM often misses them.

4.3 Why CatBoost V4 was Selected as the Production Model

CatBoost V4 was chosen despite having a lower raw accuracy than LightGBM or XGBoost. The decision is based on ROC-AUC, which is the correct metric for imbalanced binary classification tasks. Key reasons:

- LightGBM Binary achieves 78.62% accuracy by predicting the majority class almost every time — a strategy that works for accuracy but provides no real-world value. Its ROC-AUC of 0.500 confirms the model is performing at random when it comes to distinguishing High Risk from Low Risk accidents.
- CatBoost V4 achieves a ROC-AUC of 0.746, which confirms genuine class separation. The model has learned real patterns that differentiate high-risk from low-risk events.
- The optimized decision threshold of 0.5034 was applied to CatBoost V4 to improve the balance between precision and recall, resulting in 363 High Risk and 637 Low Risk classifications out of 1,000 scored records (36.3% High Risk rate).

4.4 Top Predictive Features

These are the variables that had the greatest influence on the model's predictions:

Rank	Feature	What It Represents	Importance Score
1	Distance (mi)	How far the accident affected traffic flow — longer = more severe	4,300
2	Wind_Speed_mph	Higher wind speeds correlate with more dangerous conditions	3,900
3	Year	Accident patterns changed significantly over time (2016 vs 2022)	3,400
4	Temperature_F	Extreme temperatures (very hot or very cold) raise accident risk	3,000
5	Pressure_in	Barometric pressure changes often precede severe weather events	~2,800
6	Week_of_Year	Certain weeks (winter holidays, summer) have consistently higher risk	~2,600
7	Humidity_pct	High humidity combined with cold = black ice; high heat = driver fatigue	~2,400

4.5 Forecast Model (Prophet)

 [View Prophet Forecast](#)

In addition to the classification models, a Facebook Prophet time-series model was trained on 87 months of historical accident data (2016–2023) to forecast monthly accident volumes for the next 12 months:

Metric	Value
Total Predicted Accidents (12 months)	124,000
Peak Predicted Month	December — 14,415 accidents (highest risk month)
Lowest Predicted Month	June — 6,731 accidents
Winter vs Summer	~40% more accidents in December–January vs summer months
Confidence Interval	95% confidence interval (upper and lower bounds) on all predictions

Chapter 5

Dashboard & Visualization

*Power BI Pages · KPI Highlights · Figma Design
· Screenshots*

5.1 Power BI Dashboard Pages

The final product is an 8-page interactive Power BI report, allowing users to filter by Year, State, Severity Level, Weather Type, and Rush Hour Status across all pages.

Page	Dashboard Name	Story It Tells	Most Important Number
Landing	Navigation Hub	Entry point — links to all dashboard pages with team info	500K records 7 pages 5 models
P1	Overview KPIs & Map	Scale and overall severity of US accidents	500K records 19.52% high severity
P2	Temporal Trends	When accidents happen most (hour, day, month, year)	Friday peak 202K rush hour Dec 54,589
P3	Weather Impact	Environmental conditions during accidents	177K in Fair weather — not just bad weather
P4	Geography Hotspots	Where accidents concentrate across states and cities	CA 113K Miami top city I-95 top corridor
P5	Roads Infrastructure	How road features affect accident occurrence and severity	70% with 0 road features Junction = more severe
P6	ML Results	Model comparison and feature importance	CatBoost ROC-AUC 0.746 — best classifier
P7	12-Month Forecast	Predicted future accident trends using Prophet	124K next 12 months December highest risk
P8	ML Predictions	Live model scoring on a sample of 1,000 records	36.3% High Risk Threshold 0.5034 confirmed

5.2 KPI Highlights

Across all eight dashboards, these are the four headline numbers that define the project:

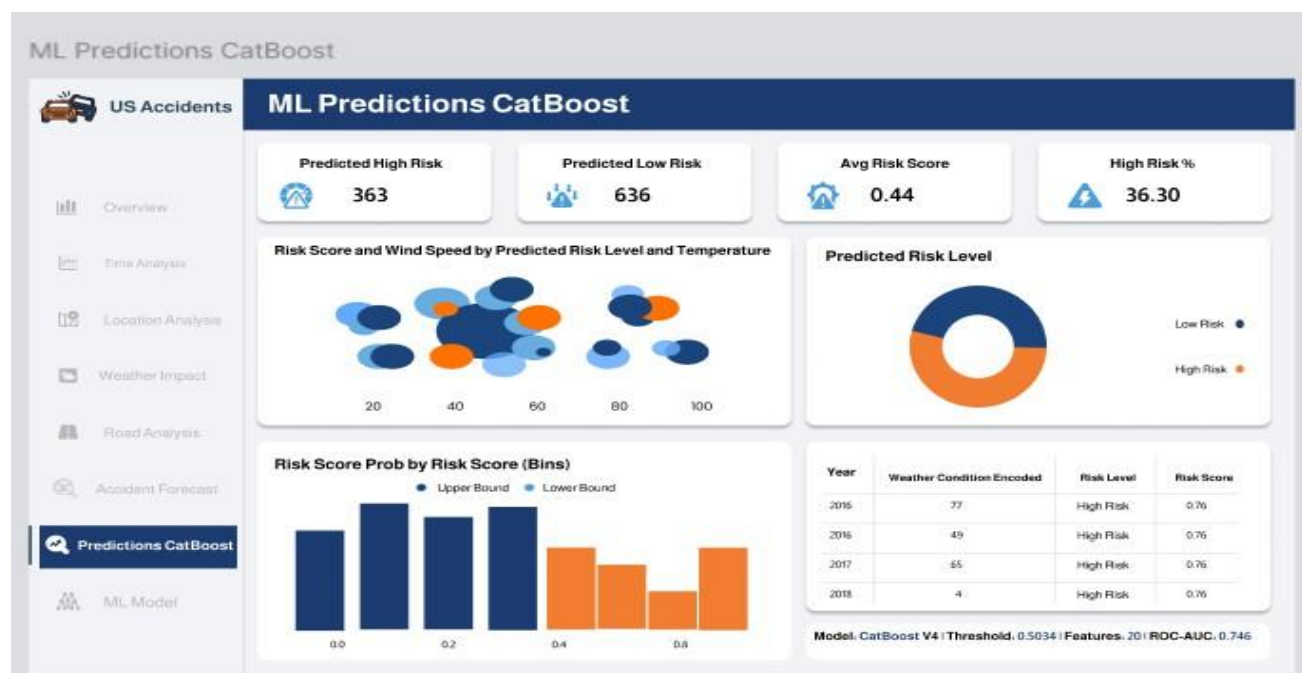
500K Total Accidents	69.83% Best Model Accuracy	202K Rush Hour Accidents	124K Predicted Next 12M
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5.3 Figma Design



[View Figma Design](#)

Before any Power BI development began, the entire dashboard was wireframed in Figma. This UI/UX design phase defined the navigation flow, colour palette (dark navy blue + orange accents), layout grid, and user interaction patterns. Using Figma allowed the team to validate the visual design with stakeholders before spending time on implementation — reducing rework and ensuring a polished, professional result.



Landing Page



Overview



Time Analysis



Location



Weather



Road



ML



Accident Forecast



5.4 Dashboard

The following pages display all eight dashboard pages from the Power BI report, showing the complete visual analysis of US road accidents data.

Dashboard 1 — Overview



[View](#)

The Overview dashboard shows the **total picture** of 500,000 US road accidents (2016–2023). At a glance, a reader can see that 80% of accidents are Moderate Impact, California leads all states with 113K accidents, and daytime accidents (275K) far outnumber night-time ones for Moderate Impact severity.



Dashboard 2 — Time Analysis



[View](#)

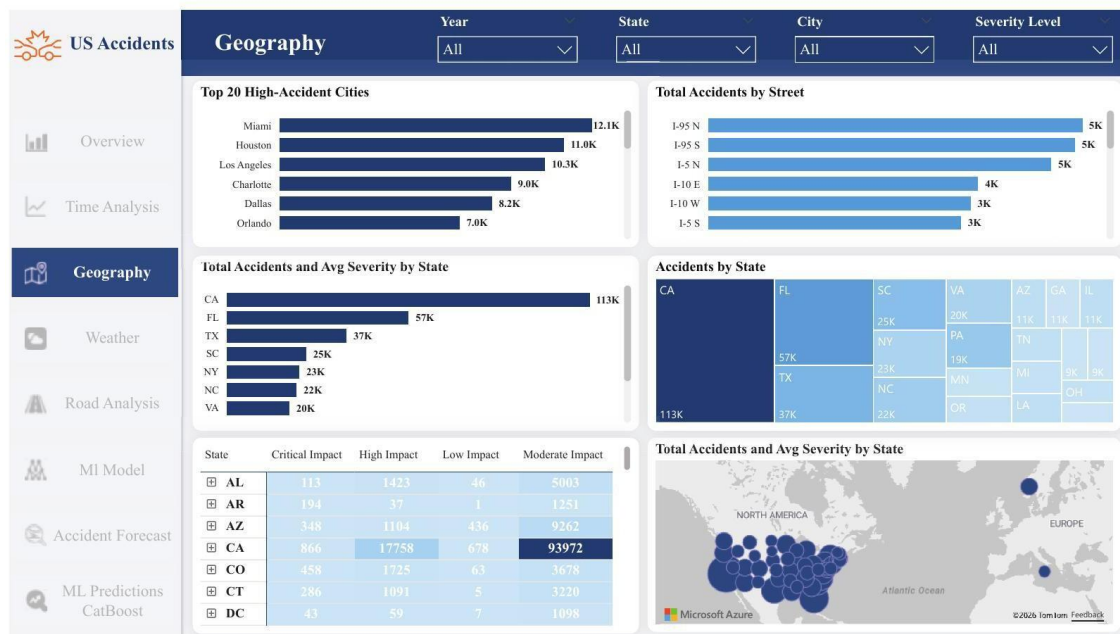
The Time Analysis dashboard answers **when accidents happen most**. Rush hour (202K accidents) and Fridays are the peak. December is consistently the highest-risk month. The hourly chart shows two clear daily spikes: morning commute (7–9 AM) and evening commute (3–6 PM).



Dashboard 3 — Geography

 [View](#)

The Geography dashboard shows **where accidents concentrate**. California (113K), Florida (57K), and Texas (37K) are the top three states. Miami leads all cities with ~12,100 accidents. Interstate I-95 North is the most accident-prone road corridor in the country.



Dashboard 4 — Weather



[View](#)

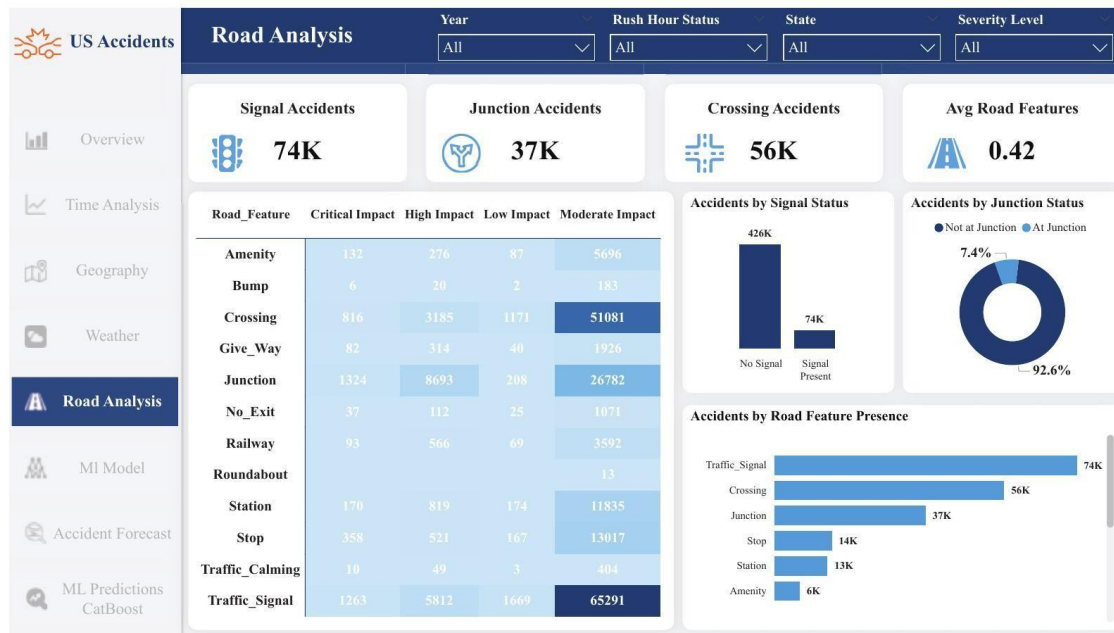
The Weather dashboard reveals a **counterintuitive finding**: the most accidents (177K) occur in 'Fair' weather — not during storms. This is because fair weather leads to faster driving and reduced caution. Extreme weather accounts for only 5% of accidents, though these tend to be more severe. Low visibility conditions still show elevated rain, snow, and fog accident counts.



Dashboard 5 — Road Analysis

 [View](#)

The Road Analysis dashboard examines **infrastructure impact**. Traffic signals (74K accidents), crossings (56K), and junctions (37K) are the highest-volume road features. Critically, 70% of all accidents occur where zero road features are present — suggesting that open road segments with no signals or crossings are high-risk zones. Junctions show the highest Critical and High Impact rates, suggesting they produce the most severe accidents.



Dashboard 6 — ML Model Results



[View](#)

The ML Results dashboard compares all trained models. **CatBoost V4 is the production model** with a ROC-AUC of 0.746 — the only model that genuinely separates High Risk from Low Risk accidents. LightGBM shows the highest raw accuracy (78.62%) but a ROC-AUC of 0.50, meaning it is no better than random at identifying dangerous accidents. The top predictive features are Distance, Wind Speed, Year, and Temperature.



Dashboard 7 — 12-Month Accident Forecast



[View](#)

The Forecast dashboard uses **Facebook's Prophet model** trained on 87 months of historical data (2016–2023) to project accident volumes for the next 12 months. The model predicts 124,000 total accidents, with December as the peak month (14,415) and June as the lowest (6,731). Winter months show 40% more accidents than summer on average. The 95% confidence interval gives reliable upper and lower bounds for planning purposes.



Dashboard 8 — ML Predictions (CatBoost)

 [View](#)

The ML Predictions dashboard scores a **sample of 1,000 records** using the production CatBoost V4 model. 363 records (36.3%) were classified as High Risk (Severity 3–4) and 637 (63.7%) as Low Risk. The average risk score is 0.44, slightly below the decision threshold of 0.5034. The scatter plot shows that high-risk accidents cluster at higher risk scores regardless of temperature, confirming the model's ability to distinguish dangerous scenarios.

